GATE ME 90 Days Challenge

Day 65/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Industrial Engineering	Operations Research	36 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on linear programming (graphical method, special cases, simplex and Big-M, duality), transportation (NWC, degeneracy, u–v method), the assignment problem (Hungarian method), project networks (CPM forward and backward pass, floats, PERT time estimates and probability, crashing) and M/M/1 queues. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

Maximise $Z = 2x + 3y$ subject to $x + y \le 8$, $x + 3y \le 18$, $2x + y \le 14$ and $x, y \ge 0$. The maximum value of Z is _____

Q2 MCQ

The LP: maximise $Z = 3x + 2y$ subject to $x - y \le 2$, $x \ge 1$ and $x, y \ge 0$ has

(A) no feasible solution

(B) an unbounded solution

(C) a unique optimal solution at (2, 0)

(D) alternative optimal solutions

Q3 MCQ

In the Big-M method, a constraint of the type $a_1x_1 + a_2x_2 \ge b$ (with $b > 0$) in a maximisation problem is converted to an equation by

(A) adding a slack variable only

(B) subtracting a surplus variable only

(C) subtracting a surplus variable and adding an artificial variable with a coefficient of $-M$ in the objective

(D) adding an artificial variable with a coefficient of $+M$ in the objective

Q4 NAT

Three warehouses S1, S2, S3 with supplies of 30, 50 and 20 units serve four shops D1–D4 with demands of 20, 40, 30 and 10 units. The unit costs (₹) to D1, D2, D3, D4 are: from S1: 3, 1, 7, 4; from S2: 2, 6, 5, 9; from S3: 8, 3, 3, 2. The total cost (in ₹) of the initial basic feasible solution found by the north-west corner rule is _____

Q5 MCQ

An initial basic feasible solution of an m × n transportation problem has fewer than m + n − 1 positive allocations. Before applying the u–v (MODI) method, one should

(A) drop one row of the cost table

(B) add a dummy row of zero cost

(C) declare the solution optimal

(D) allocate a very small quantity ε to a suitable empty cell that does not form a closed loop with the occupied cells

Q6 NAT

Three workers W1, W2, W3 are to be assigned one each to jobs J1, J2, J3. The costs (₹) are W1: 8, 4, 7; W2: 5, 2, 3; W3: 9, 4, 8 (for J1, J2, J3). The minimum total cost (in ₹) is _____

Q7 MCQ

To solve an assignment problem in which the entries are profits to be maximised by the Hungarian method, the profit matrix is first

(A) converted to a loss matrix by subtracting every entry from the largest entry of the matrix

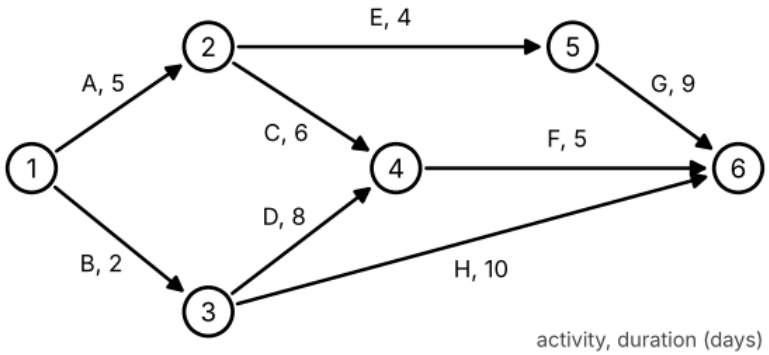
(B) multiplied by the number of rows

(C) replaced by its transpose

(D) divided by the largest entry, and then the method is applied unchanged to maximise

Q8 NAT

The activity-on-arrow network of a project is shown, with durations in days. The project duration (in days) is _____



Q9 NAT

For the project network of the previous question, the total float of activity D (3–4), in days, is _____

Q10 MCQ

An activity has optimistic, most likely and pessimistic times of 3, 6 and 15 days. Its expected time and standard deviation are

(A) 8 days and 2 days

(B) 7 days and 2 days

(C) 7 days and 4 days

(D) 6 days and 2 days

Q11 NAT

The expected length of the critical path of a project is 50 days and the variance of the project duration is 25 days². Assuming a normal distribution (z = 1.645 for 95 % probability), the due date (in days) that gives a 95 % chance of completion (round off to one decimal place) is _____

Q12 MCQ

While crashing a project to reduce its duration at minimum extra cost, the activity crashed first is the one

(A) with the longest normal duration in the network

(B) with the least cost slope among all activities

(C) on the critical path with the least cost slope

(D) on a non-critical path with the largest float

Q13 NAT

Customers arrive at a single-window counter in a Poisson stream at 6 per hour, and the service time is exponential with a mean of 7.5 minutes (M/M/1). The probability that there are at least 3 customers in the system (round off to three decimal places) is _____

Q14 MCQ

In an M/M/1 queue the arrival rate is 3 per hour and the service rate is 4 per hour. The probability that the server is idle and the average number of customers in the queue (excluding the one in service) are

(A) 0.75 and 2.25

(B) 0.25 and 3

(C) 0.75 and 3

(D) 0.25 and 2.25

Q15 MCQ

A primal LP has 3 constraints (other than non-negativity) and 2 decision variables. Its dual has

(A) 3 variables and 2 constraints

(B) 2 variables and 3 constraints

(C) 3 variables and 3 constraints

(D) 2 variables and 2 constraints

Answer Key

Q	Type	Answer
Q1	NAT	21
Q2	MCQ	(B) an unbounded solution
Q3	MCQ	(C) subtracting a surplus variable and adding an artificial variable with a coefficient of −M in the objective
Q4	NAT	400
Q5	MCQ	(D) allocate a very small quantity ε to a suitable empty cell that does not form a closed loop with the occupied cells
Q6	NAT	15
Q7	MCQ	(A) converted to a loss matrix by subtracting every entry from the largest entry of the matrix
Q8	NAT	18
Q9	NAT	3
Q10	MCQ	(B) 7 days and 2 days
Q11	NAT	58.2 (accept 58.1 to 58.3)
Q12	MCQ	(C) on the critical path with the least cost slope
Q13	NAT	0.422 (accept 0.418 to 0.426)
Q14	MCQ	(D) 0.25 and 2.25
Q15	MCQ	(A) 3 variables and 2 constraints

Detailed Solutions

Q1 Answer: 21

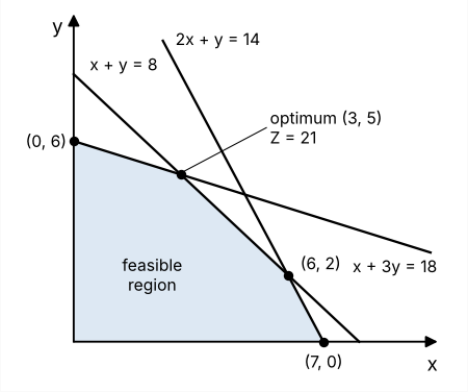
Concept: The optimum of an LP lies at a corner (extreme point) of the feasible region; evaluate Z at every corner.

Corners: (0, 0); (7, 0) on $2x + y = 14$; (6, 2) where $2x + y = 14$ meets $x + y = 8$; (3, 5) where $x + y = 8$ meets $x + 3y = 18$; (0, 6)

Z = 0, 14, 18, 21, 18

Maximum Z = **21** at x = 3, y = 5

Exam tip: Check each intersection against the third constraint: $x + 3y = 18$ and $2x + y = 14$ meet at (4.8, 4.4), which violates $x + y \le 8$, so it is not a corner.



Q2 Answer: (B) an unbounded solution

Concept: If the feasible region is open in a direction in which Z keeps increasing, the LP is unbounded.

Take $x = 1$ and let y grow: $x - y = 1 - y \leq 2$ holds for every $y \geq 0$, so all points (1, y) are feasible

$Z = 3 + 2y \to \infty$: the solution is **unbounded (B)**

Exam tip: In the simplex table, unboundedness shows up as an entering column with no positive entry (no ratio can be formed).

Q3 Answer: (C) subtracting a surplus variable and adding an artificial variable with a coefficient of -M in the objective

Concept: A surplus variable turns ≥ into =, but it enters with −1, so it cannot form the initial identity basis.

An artificial variable A is added to give a starting basic feasible solution; to drive it out, it is penalised with −M (very large) in a max problem (+M in a min problem): **(C)**

If an artificial variable stays positive in the final table, the LP is infeasible.

Exam tip: The two-phase method does the same job: phase 1 minimises the sum of artificial variables.

Q4 Answer: 400

Concept: NWC rule: start at the top-left cell, allocate the most possible, and move right (demand left over) or down (supply left over).

S1–D1 = 20 (S1 left 10); S1–D2 = 10 (S1 done); S2–D2 = 30; S2–D3 = 20 (S2 done); S3–D3 = 10; S3–D4 = 10

Cost = 20 × 3 + 10 × 1 + 30 × 6 + 20 × 5 + 10 × 3 + 10 × 2 = 60 + 10 + 180 + 100 + 30 + 20 = **₹400**

Six allocations = m + n − 1 = 3 + 4 − 1, so the solution is non-degenerate.

Exam tip: NWC ignores costs; VAM or the least-cost method usually starts much closer to the optimum.

Q5 Answer: (D) allocate a very small quantity ε to a suitable empty cell that does not form a closed loop with the occupied cells

Concept: The u–v method needs m + n − 1 occupied (basic) cells to solve for all the u_i and v_j; fewer cells means the solution is degenerate.

Put ε (→ 0) in an independent empty cell, preferably a low-cost one, so that no closed loop is formed: **(D)**. Then proceed normally and set ε = 0 at the end.

Exam tip: A dummy row or column is used for an unbalanced problem (total supply ≠ total demand), not for degeneracy.

Q6 **Answer: 15**

Concept: Hungarian method: subtract row minima, then column minima, and make an assignment using independent zeros.

Row reduction: W1: 4, 0, 3; W2: 3, 0, 1; W3: 5, 0, 4. Column reduction (minima 3, 0, 1): W1: 1, 0, 2; W2: 0, 0, 0; W3: 2, 0, 3

Zeros can be covered by 2 lines (< 3), so subtract the smallest uncovered value 1 and add it at line intersections: W1: 0, 0, 1; W2: 0, 1, 0; W3: 1, 0, 2

Assign W3–J2, W1–J1, W2–J3: 8 + 3 + 4 = **₹15**

Exam tip: With only 3! = 6 possibilities, quickly listing them (15, 16, 16, 17, 18, 18) is a fast check in the exam.

Q7 **Answer: (A) converted to a loss matrix by subtracting every entry from the largest entry of the matrix**

Concept: The Hungarian method minimises. Maximising P is the same as minimising ($P_{max} - P$) (or $-P$).

Subtract each entry from the largest one to get an opportunity-loss matrix, then minimise: **(A)**

The optimal assignment is the same; the total profit is read from the original matrix.

Exam tip: An unbalanced problem (more jobs than workers) needs a dummy row with zero entries first.

Q8 **Answer: 18**

Concept: Forward pass: earliest time of an event E_j = max over incoming activities of ($E_i + t_{ij}$); the project time is E of the end event.

E1 = 0, E2 = 5, E3 = 2, E4 = max(5 + 6, 2 + 8) = 11, E5 = 5 + 4 = 9

E6 = max(11 + 5, 9 + 9, 2 + 10) = max(16, 18, 12) = **18 days**; the critical path is A–E–G (1–2–5–6)

Exam tip: Path lengths: A–C–F 16, B–D–F 15, A–E–G 18, B–H 12. Listing all paths is a quick check in small networks.

Q9 **Answer: 3**

Concept: Total float = LS − ES = $L_j - E_i - t_{ij}$, found from a backward pass.

Backward pass: L6 = 18, L5 = 9, L4 = 18 − 5 = 13, L2 = min(13 − 6, 9 − 4) = 5, L3 = min(13 − 8, 18 − 10) = 5

TF(D) = L4 − E3 − 8 = 13 − 2 − 8 = **3 days**

Free float of D = E4 − E3 − 8 = 11 − 10 = 1 day.

Exam tip: TF of an activity = (critical path length) − (length of the longest path through that activity) = 18 − 15 = 3.

Q10 **Answer: (B) 7 days and 2 days**

Concept: PERT (beta distribution): $t_e = (a + 4m + b)/6$, $\sigma = (b - a)/6$.

$t_e = (3 + 24 + 15)/6 = 42/6 = 7$ days; $\sigma = (15 - 3)/6 = 2$ days: **(B)**

Exam tip: t_e lies on the side of the longer tail (b − m = 9 > m − a = 3), so it exceeds the most likely time.

Q11 **Answer: 58.2 (accept 58.1 to 58.3)**

Concept: $T = T_e + z\sigma$, where σ is the square root of the sum of the variances of the critical activities.

$\sigma = \sqrt{25} = 5$ days

$T = 50 + 1.645 \times 5 = 50 + 8.225 =$ **58.2 days**

Exam tip: Never add the standard deviations of activities; add variances and then take the square root.

Q12 **Answer: (C) on the critical path with the least cost slope**

Concept: Only critical activities shorten the project; cost slope = (crash cost − normal cost)/(normal time − crash time).

Choose the critical activity with the least cost slope: **(C)**. When several paths become critical, all of them must be shortened together (choose the cheapest combination).

Stop when the extra direct cost per day exceeds the indirect cost saved per day: that gives the optimum duration.

Exam tip: Crashing a non-critical activity only adds cost, it never shortens the project.

Q13 **Answer: 0.422 (accept 0.418 to 0.426)**

Concept: In an M/M/1 queue $P_n = (1 - \rho)\rho^n$, so $P(N \geq k) = \rho^k$.

$\mu = 60/7.5 = 8$ per hour, $\rho = \lambda/\mu = 6/8 = 0.75$

$P(N \geq 3) = 0.75^3 =$ **0.422**

Exam tip: P(an arriving customer has to wait) = $P(N \geq 1) = \rho = 0.75$.

Q14 **Answer: (D) 0.25 and 2.25**

Concept: $P_0 = 1 - \rho$; $L_q = \rho^2/(1 - \rho) = \lambda^2/[\mu(\mu - \lambda)]$.

$\rho = 3/4 = 0.75$, so $P_0 = 0.25$

$L_q = 0.5625/0.25 = 2.25$; $L = \rho/(1 - \rho) = 3$: **(D)**

Exam tip: $L - L_q = \rho$ always, because the server holds on average ρ customers.

Q15 **Answer: (A) 3 variables and 2 constraints**

Concept: One dual variable for each primal constraint and one dual constraint for each primal variable; the coefficient matrix is transposed.

3 primal constraints → 3 dual variables; 2 primal variables → 2 dual constraints: **(A)**

The optimal dual variables are the shadow prices of the primal resources.

Exam tip: The dual of the dual is the primal; a primal with many constraints and few variables is often easier to solve through its dual.

What next?

Day 65 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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